

# McKell Woodland

[mckell.woodland@gmail.com](mailto:mckell.woodland@gmail.com)

[github.com/mckellwoodland](https://github.com/mckellwoodland)

[mckellwoodland.github.io](https://mckellwoodland.github.io)

---

## Education

---

**Rice University**, Houston, TX

Aug 2024

Ph.D. Computer Science

**Brigham Young University**, Provo, UT

Apr 2018

B.S. Applied and Computational Mathematics

---

## Professional Experience

---

**The University of Texas MD Anderson Cancer Center**, Houston, TX

Jan 2020 – Present

*Data Scientist; Graduate Research Assistant*

- Researching out-of-distribution detection and generative modeling evaluation in medical imaging
- Peer-reviewed publications: MELBA, MICCAI 2024, UNSURE 2023, Diagnostics, SASHIMI 2022, & Eur. J. Radiol.
- Awards: Outstanding Reviewer – Honorable Mention (MICCAI 2024 & 2025), Best Spotlight Paper (UNSURE 2023), Graduate Student Travel Grant, Young Investigator Award – Significance (PBDW 2021), and Worldwide Innovations in Medical Physics Scholarship (WIMP 2022)
- Invited talks: RSNA 2025, TMI Symposium, BYU Computer Science, MSB Thesis Madness (MICCAI 2024), UNSURE 2023, AAPM 2021-2023, & MedPhys Slam (SWAAPM 2022)
- Participated in a public debate at PBDW 2023; Spoke on a MICCAI Student Board panel
- Moderated session at 2<sup>nd</sup> Annual Research Retreat in Oncological Data and Computational Sciences

**Microsoft**, Redmond, WA (remote)

May 2020 - Aug 2020

*Data Science Intern*

- Devised a similarity score in Python to analyze overlap between suspicious behavior detectors
- Produced, using Kusto, a tool for Microsoft Threat Experts to use the score to detect malicious behavior
- Moderated a Q&A session with the TF and CTO of Microsoft's AI Cognitive Services

**Lawrence Livermore National Laboratory**, Livermore, CA

May 2019 - Aug 2019

*Data Science Research and Development Intern*

- Developed an event detection model for videos by embedding an R2U-Net into a ConvLSTM using PyTorch
- Crafted 100,000 colorful moving MNIST with obscurations videos with OpenCV (VRAM)
- Gave final model (99% acc on VRAM) to client to apply to classified data with obscured & moving objects

**Perception Control and Cognition Lab**, Provo, UT

Jan 2018 – May 2019

*Graduate Research Assistant*

- Trained a Rainbow DQN on game states embedded by a CVLAE to reduce sample complexity in RL
- Awards: Session Winner (BYU SRC 2019); Invited Talk: BYU SRC 2019

**3M Health Information Systems**, Murray, UT

May 2018 – Aug 2018

*Data Science Intern*

- Made an RNN on an EC2 instance to predict over 50,000 diagnoses from 1.7 million documents using Keras
- Built the basis of a deep learning framework in Keras to predict DRG codes
- Assembled a framework to analyze the text behind any amount of medical notes

**National Security Agency**, Laurel, MD

May 2017 – Aug 2017

*Director's Summer Program Intern*

- Implemented custom convolutional neural nets using Python and received TS/SCI clearance
- Analyzed an inertial navigation problem with noisy data and submitted an internal technical paper
- Prepared a briefing for the Deputy Director and presented findings to the IDA at Princeton

**Brigham Young University**, Provo, UT

Jan 2016– Dec 2017

*Research Assistant; Developer*

- Researched optimal facility location, developable conical mechanisms, and instructional technology
- Peer-reviewed publications: J. Mech. Robot, JSSR, & IRRODL; Invited talks: IDETC/CIE 2020, NERCCS 2019, & BYU SRC 2018
- Utilized Python, Mathematica, & SQL technologies

---

## Other Research Experience

---

### Texas Children's Hospital; Medical Informatics Corp, Houston, TX

Aug 2019 – Dec 2019

- Created a novel classification algorithm to detect the onset of a heart rhythm disorder with 90% AUROC
- Engineered a novel waveform visualization tool (using t-SNE) for use in a development pipeline
- Team took first place in Rice's Data2Knowledge Showcase; poster presentation in WFPICCS 2020

### Computer Science Department, BYU, Provo, UT

Aug 2017– Dec 2017

- Extracted 23 features from web-scraped BYU speeches and used a wrapper algorithm for feature selection
- Ran 4 machine learning models on the features and achieved 79% accuracy on the speeches' popularity
- Invited talks: NCUR 2018

---

## Peer-Reviewed Articles

---

- **Woodland, M.E.**, Altaie, M., O'Connor, C.S., Castelo, A.H., Lebimoyo, O.C., Gupta, A.C., Yung, J.P., Kinahan, P.E., Fuller, C.D., Koay, E.J., Odisio, B.C., Patel, A.B., & Brock, K.K. (2025). Generative Modeling for Interpretable Anomaly Detection in Medical Imaging: Applications in Failure Detection and Data Curation. *Bioengineering*.
- **Woodland, M.**, Patel, N., Castelo, A., Al Taie, M., Eltaher, M., Yung, J.P., Netherton, T.J., Calderone, T.L., Sanchez, J.I., Cleere, D.W., Elsaiey, A., Gupta, N., Victor, D., Beretta, L., Patel, A.B., & Brock, K.K. (2024). Dimensionality Reduction and Nearest Neighbors for Improving Out-of-Distribution Detection in Medical Image Segmentation. *Machine Learning for Biomedical Imaging*, 2(UNSURE2023 special issue), 2006-2052. [doi.org/10.59275/j.melba.2024-g93a](https://doi.org/10.59275/j.melba.2024-g93a)
- Baroudi, H., Brock, K.K., Cao, W., Chen, X., Chung, C., Court, L.E., El Basha, M.D., Farhat, M., Gay, S., Gronberg, M.P., Gupta, A.C., Hernandez, S., Huang, K., Jaffray, D.A., Lim, R., Marquez, B., Nealon, K., Netherton, T.J., Nguyen, C.M., Reber, B., Rhee, D.J., Salazar, R.M., Shanker, M.D., Sjogreen, C., **Woodland, M.**, Yang, J., Yu, C., & Zhao, Y. (2023). Automated Contouring and Planning in Radiation Therapy: What Is 'Clinically Acceptable'? *Diagnostics*, 13, 667. [doi.org/10.3390/diagnostics13040667](https://doi.org/10.3390/diagnostics13040667)
- Sen, A., Troncoso, P., Venkatesan, A., Pagel, M.D., Nijkamp, J.A., He, Y., Lesage, A., **Woodland, M.**, & Brock, K.K. (2021). Correlation of In-Vivo Imaging with Histopathology: A Review. *European Journal of Radiology*, 144, 109964. [doi.org/10.1016/j.ejrad.2021.109964](https://doi.org/10.1016/j.ejrad.2021.109964)
- **Woodland, M.**, Hsiung, M., Matheson, E., Safsten, A., Greenwood, J., Halverson, D.M., & Howell, L.L. (2021). Analysis of the Rigid Motion of a Developable Conical Mechanism. *Journal of Mechanisms and Robotics*, 13, 031106-1 to 031106-8. [doi.org/10.1115/1.4050294](https://doi.org/10.1115/1.4050294)
- Kimmons, R., Hunsaker, E., Jones, E.J., & **Stauffer, M.** (2019). The nationwide landscape of K-12 school websites in the United States: Systems, services, intended audiences, and adoption patterns. *International Review of Research in Open and Distributed Learning*, 20(3). [doi.org/10.19173/irrodl.v20i4.3794](https://doi.org/10.19173/irrodl.v20i4.3794)
- Kimmons, R., McGuire, K., **Stauffer, M.**, Jones, E.J., Gregson, M., & Austin, M. (2017). Religious identity, expression, and civility in social media: Results of data mining Latter-day Saint Twitter accounts. *Journal for the Scientific Study of Religion*, 56(3), 180-210. [doi.org/10.1111/jssr.12358](https://doi.org/10.1111/jssr.12358)

---

## Conference Proceedings

---

- **Woodland, M.**, Castelo, A., Al Taie, M., Albuquerque Marques Silva, J., Eltaher, M., Mohn, F., Kundu, S., Yung, J.P., Patel, A.B., & Brock, K.K. (2024). Feature Extraction for Generative Medical Imaging Evaluation: New Evidence Against an Evolving Trend. *Medical Image Computing and Computer Assisted Intervention 2024*.
- **Woodland, M.**, Patel, N., Al Taie, M., Yung, J.P., Netherton, T.J., Patel, A.B., & Brock, K.K. (2023). Dimensionality Reduction for Improving Out-of-Distribution Detection in Medical Image Segmentation. In: Sudre, C.H., Baumgartner, C.F., Dalca, A., Mehta, R., Qin, C., Wells, W.M. (eds) *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging 2023*. LNCS, vol 14291. Springer, Cham. [doi.org/10.1007/978-3-031-44336-7\\_15](https://doi.org/10.1007/978-3-031-44336-7_15)
- **Woodland, M.**, Wood, J., Anderson, B.M., Kundu, S., Lin, E., Koay, E., Odisio, B., Chung, C., Kang, H.C., Venkatesan, A.M., Yedururi, S., De, B., Lin, Y.-M., Patel, A.B., & Brock, K.K. (2022). Evaluating the Performance of StyleGAN2-ADA on Medical Images. In: Zhao, C., Svoboda, D., Wolterink, J.M., Escobar, M. (eds) *Simulation and Synthesis in Medical Imaging 2022*. LNCS, vol 13570. Springer, Cham. [doi.org/10.1007/978-3-031-16980-9\\_14](https://doi.org/10.1007/978-3-031-16980-9_14)

- **Woodland, M.**, Hsiung, M., Matheson, E.L., Safsten, C.A., Greenwood, J., Halverson, D.M., & Howell, L.L. (2020). Analysis of the Rigid Motion of a Developable Conical Mechanism. *International Design Engineering Technical Conferences & Computers and Information in Engineering 2020*. [doi.org/10.1115/DETC2020-22643](https://doi.org/10.1115/DETC2020-22643)
- Henriksen, A., Jafek, B., O'Bryant, J., & **Stauffer, M.** (2018). Brigham Young University Speeches Popularity Predictor. *National Conference on Undergraduate Research 2018*. [libjournals.unca.edu/ncur](http://libjournals.unca.edu/ncur)

---

## Abstracts

---

- **Woodland, M.**, O'Connor, C., Wood, J., Patel, A.B., & Brock, K.K. (2023). Interpretable Out-of-Distribution Detection with Generative Adversarial Networks. *Medical Physics*, 50(6), e154-155. [doi.org/10.1002/mp.16525](https://doi.org/10.1002/mp.16525)
- **Woodland, M.**, Wood, J., O'Connor, C., Patel, A.B., & Brock, K.K. (2022). StyleGAN2-based Out-of-Distribution Detection for Medical Imaging. In: *Medical Imaging meets NeurIPS 2022*. [sites.google.com/view/med-neurips-2022/abstracts](https://sites.google.com/view/med-neurips-2022/abstracts)
- Reber, B., van Dijk, L.V., Anderson, B.M., Mohamed, A.S., Rigaud, B., He, Y., **Woodland, M.**, Fuller, C.D., Lai, S.Y., and Brock, K.K. (2022) Comparison of Machine Learning and Deep Learning Methods for the Prediction of Osteoradionecrosis Resulting from Head and Neck Cancer Radiation Therapy. *International Journal of Radiation Oncology Biology Physics*, 114(3), e124. [doi.org/10.1016/j.ijrobp.2022.07.946](https://doi.org/10.1016/j.ijrobp.2022.07.946)
- **Woodland, M.**, Wood, J., Anderson, B.M., Kundu, S., Lin, E., Koay, E., Odisio, B., Chung, C., Kang, H.C., Venkatesan, A., Yedururi, S., De, B., Lin, Y., Patel, A.B., & Brock, K.K. (2022). Comparing Transfer Learning, Data Augmentation, and Data Expansion in the Improvement of Medical Image Generation. *Medical Physics*, 49(6), 134. [doi.org/10.1002/mp.15769](https://doi.org/10.1002/mp.15769)
- **Woodland, M.**, Patel, A.B., Anderson, B.M., Lin, E., Koay, E., Odisio, B., & Brock, K.K. (2021). GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation. *Medical Physics*, 48(6). [doi.org/10.1002/mp.15041](https://doi.org/10.1002/mp.15041)
- Reber, B., Anderson, B.M., Mohamed, A., Van Dijk, L., Rigaud, B., McCulloch, M., He, Y., **Woodland, M.**, Fuller, C., Lai, S., & Brock, K.K. (2021). Predicting Osteoradionecrosis from Head and Neck Radiotherapy Using a Residual Convolutional Neural Network. *Medical Physics*, 48(6). [doi.org/10.1002/mp.15041](https://doi.org/10.1002/mp.15041)
- Sailsbery, M., Heiner, J., Robertson, C., **Stauffer, M.**, & Jarvis, T. (2019). Facility Location Using Markov Chains on Spatial Networks. In: *Northeast Regional Conference on Complex Systems 2019*. [coco.binghamton.edu/nerccs](http://coco.binghamton.edu/nerccs)

---

## Invited Talks

---

- *Radiological Society of North America 2025.*
- Predicting Segmentation Model Failure with Out-of-Distribution Detection and Generative Adversarial Networks. *Tumor Measurement Initiative Symposium 2025.*
- Predicting Liver Segmentation Model Failure with Feature-based Out-of-Distribution Detection and Generative Adversarial Networks. *Computer Science Colloquia at Brigham Young University, 2024.*
- Predicting Liver Segmentation Model Failure with Feature-Based Out-of-Distribution Detection and Generative Adversarial Networks. *MICCAI Student Board PhD Thesis Madness 2024.*
- Dimensionality Reduction for Improving Out-of-Distribution Detection in Medical Image Segmentation. *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging 2023.*
- Interpretable Out-of-Distribution Detection with Generative Adversarial Networks. *American Association of Physicists in Medicine 2023.*
- Comparing Transfer Learning, Data Augmentation, and Data Expansion in the Improvement of Medical Image Generation. *American Association of Physicists in Medicine 2022.*
- Houston, Our AI Models Have a Problem! *Southwest Regional Chapter of the American Association of Physicists in Medicine 2022.*
- GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation. *American Association of Physicists in Medicine 2021.*
- Understanding LDS Twitter use. *Instructional Design and Learning Community Conference 2016.*

---

## Posters

---

- Feature Extraction for Generative Medical Imaging Evaluation: New Evidence Against an Evolving Trend. *Medical Imaging Computing and Computer Assisted Intervention 2024.*

- Dimensionality Reduction for Improving Out-of-Distribution Detection in Medical Image Segmentation. *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging 2023*.
- Interpretable OOD Detection using GANs: An Application to the MIDRC Data Repository. *Practical Big Data Workshop 2022*.
- StyleGAN2-based Out-of-Distribution Detection for Medical Imaging. *Medical Imaging meets NeurIPS 2022*.
- Evaluating the Performance of StyleGAN2-ADA on Medical Images. *Simulation and Synthesis in Medical Imaging 2022*.
- StyleGAN2-based Out-of-Distribution Detection in Computed Tomography. *Practical Big Data Workshop 2022*.
- Improving the Generation of Synthetic Medical Images using Data Augmentation and Transfer Learning. *Southwest Regional Chapter of the American Association of Physicists in Medicine 2022*.
- Detecting Out-of-Distribution Images for Active Learning using a Generative Adversarial Network (StyleGAN2). *Practical Big Data Workshop 2021*.
- Comparison of Machine Learning Methods for the Prediction of Osteoradionecrosis (ORN) from Head and Neck Cancer (HNC) Radiation Therapy. *Practical Big Data Workshop 2021*.
- GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation. *Winter Institute of Medical Physics 2021*.
- A Novel Algorithm for Early Detection of Junctional Ectopic Tachycardia in Patients with Congenital Heart Disease. *World Federation of Pediatric Intensive & Critical Care Societies 2020*.

---

## Panel Participant

---

- Panel Discussion on the trips and tips of MICCAI Review, Rebuttal, Challenge, and Workshop. *MICCAI Student Board Webinar Series, 2024*.
- "And You're Out!" – Setting Minimum Expectations for Big Data/AI Abstracts. *Practical Big Data Workshop 2023*.

---

## Awards

---

(2025) Outstanding Reviewer – Honorable Mention. *Medical Image Computing and Computer Assisted Intervention 2025*.

(2024) Outstanding Reviewer – Honorable Mention. *Medical Image Computing and Computer Assisted Intervention 2024*.

(2023) Best Spotlight Paper. *Uncertainty for the Safe Utilization of Machine Learning in Medical Imaging 2023*.

(2023) Graduate Student Travel Grant. *Rice Engineering Alumni*.

(2021) Early Career Investigator Significance Award. *Practical Big Data Workshop 2021*.

(2021) Worldwide Innovations in Medical Physics Finalist. *Winter Institute of Medical Physics 2021*.

(2019) Showcase Winner. *Rice Data2Knowledge Showcase*.

(2019) Session Winner. *Brigham Young University Student Research Conference*.

(2019) Y-Serve Exceptional Leader. *Brigham Young University Y-Serve*.

(2018) Outstanding Mathematician. *Brigham Young University Math Department*.

(2017) Outstanding Mathematician. *Brigham Young University Math Department*.

---

## Technical Skills

---

**Languages:** Python, C++, Java, HTML, and C

**Platforms:** Docker, Kubernetes, Linux, GitHub, Amazon Web Services, Spark, SQL, Kusto, and LaTeX

**Python Packages:** PyTorch, TensorFlow, PySpark, Scikit-Learn, Keras, OpenCV, Autograd, and Pandas

---

## Teaching Experience

---

**The Coding School**, Remote

Aug 2019 – Present

*Lead Instructor; Developer; Young Professional Advisory Board Member*

- Taught introductory Python and AI courses to over 400 elementary, middle, and high school students
- Instructed 300+ HBCU and community college professors on how to introduce AI into their classrooms
- Tutored 5 students one-on-one in data structures, OOP, algorithms, and AI in both Java and Python
- Authored the deep learning and SQL curriculum and edited the machine learning curriculum
- Aided in the design of the governance structure of the Young Professional Advisory Board

**Computer Science, Brigham Young University - Idaho, Remote**

Apr 2022 – July 2022

*Online Adjunct Instructor*

- Taught the online pilot course CSE 450: Machine Learning and Data Mining
- Served as the Online Course Representative for CSE 450
- Received 12% higher reviews than all other online instructors in terms of concern, timeliness, & enthusiasm

**Computer Science, Rice University, Houston, TX***Teaching Assistant; Instructional Coach*

Oct 2020 – Apr 2022

- TA'd: Statistics for Computing and Data Science (Fall 2021), Introduction to Computer Security (Spring 2021), and Programming for Data Science (Fall 2020)
- Help students prepare to give a research presentation in the department's Graduate Research Seminar

**Mathematics, Brigham Young University, Provo, UT**

Jan 2017 – Dec 2017

*Curriculum Developer*

- Wrote 7 Python labs for the Applied & Computational Mathematics curriculum: iterative solvers, profiling, SymPy, SQL 1 & 2, one-dimensional optimization, & gradient descent
- Worked with collaborators to edit over 15 different labs

---

## Academic Services

---

**Reviewer**, Applied Sciences, 2025**Reviewer**, Medical Image Computing & Computer Assisted Intervention, 2024-2025**Reviewer**, Communications Medicine, 2024**Reviewer**, Uncertainty for Safe Utilization of Machine Learning in Medical Imaging, 2024**Reviewer**, Workshop on Deep Generative Models for Health held in conjunction with NeurIPS, 2023

---

## Community Involvement

---

**Rice University, Houston, TX**

Aug 2019 – Aug 2024

*Panelist, Mentor, Teaching Assistant*

- Spoke on a graduate student panel for the Rice CS/IO club
- Mentored undergraduate students in the 2022 Rice Datathon
- Tutored international online master's students who were struggling with the data science curriculum

**Summer STEM Institute, Remote**

Jun 2021 – Aug 2021

*Research Mentor*

- Advised a high-school student who built a glioblastoma detector on MRI scans using a pre-trained CNN
- Guided another student who built an arrhythmia detector on ECG signals using a pre-trained ResNet
- Delivered a talk on deep learning in medical imaging to ~300 high school students

**BYU Y-Serve: Kids Who Code, Provo, UT**

Aug 2018 - Apr 2019

*Executive Director*

- Established a program that facilitates university students teaching kids in the local community coding skills
- Facilitated a Girls Who Code club (20+ girls) at an elementary school (taught Scratch)
- Received a top leadership award from the Y-Serve organization

**Utah STEM Action Center, Salt Lake City, UT**

Apr 2018 – May 2019

*Ambassador*

- Promoted STEM at 10+ events and called 100+ schools to teach them about STEM opportunities
- Spoke on a panel at a "Girls Who Code" event at Adobe to 200+ middle school girls
- Accepted certificate from Utah Governor Gary Herbert for service

**BYU Women in Computer Science, Provo, UT**

Jan 2018 – Dec 2018

*Public Outreach Officer*

- Planned a hackathon to help women at BYU have projects for interviews
- Interacted with companies to promote opportunities among the female computer science students
- Increased the number of social media followers by 500%